Lecture 28 Principal Component Analysis II

STAT 441: Statistical Methods for Learning and Data Mining

Wenlu TANG

Department of Mathematical and Statistical Sciences
University of Alberta

Winter, 2026

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Outline

1. Illustration
2. Proportion Variance Explained
3. Summary and Remark

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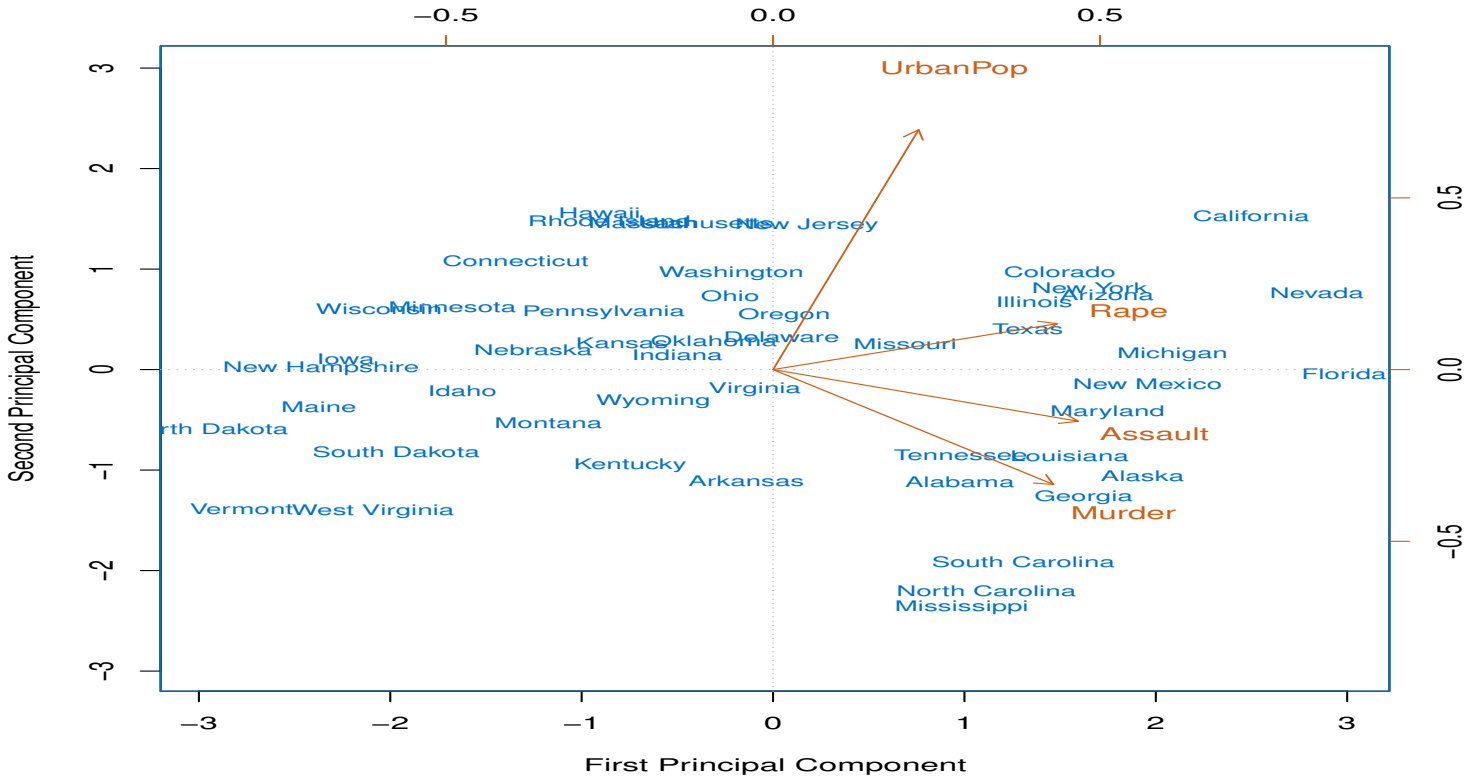
Illustration

- `USAarrests` data: For each of the fifty states in the United States, the data set contains the number of arrests per $100,000$ residents for each of three crimes: `Assault`, `Murder`, and `Rape`. We also record `UrbanPop` (the percent of the population in each state living in urban areas).
- The principal component score vectors have length $n = 50$, and the principal component loading vectors have length $p = 4$.
- PCA was performed after standardizing each variable to have mean zero and standard deviation one.

PCA plot

The first two principal components for the `USArrests` data.



PCA plot

- The blue state names represent the scores for the first two principal components.
- The orange arrows indicate the first two principal component loading vectors (with axes on the top and right).
- For example, the loading for `Rape` on the first component is 0.54, and its loading on the second principal component 0.17 [the word `Rape` is centered at the point $(0.54, 0.17)$].
- This figure is known as a biplot, because it displays both the principal component scores and the principal component loadings.

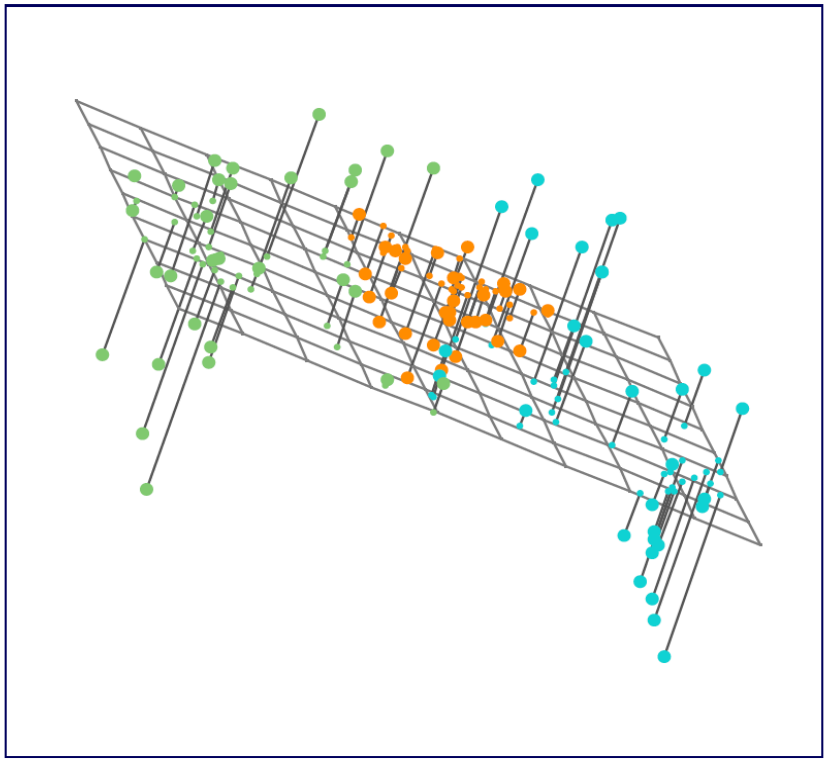
PCA loadings

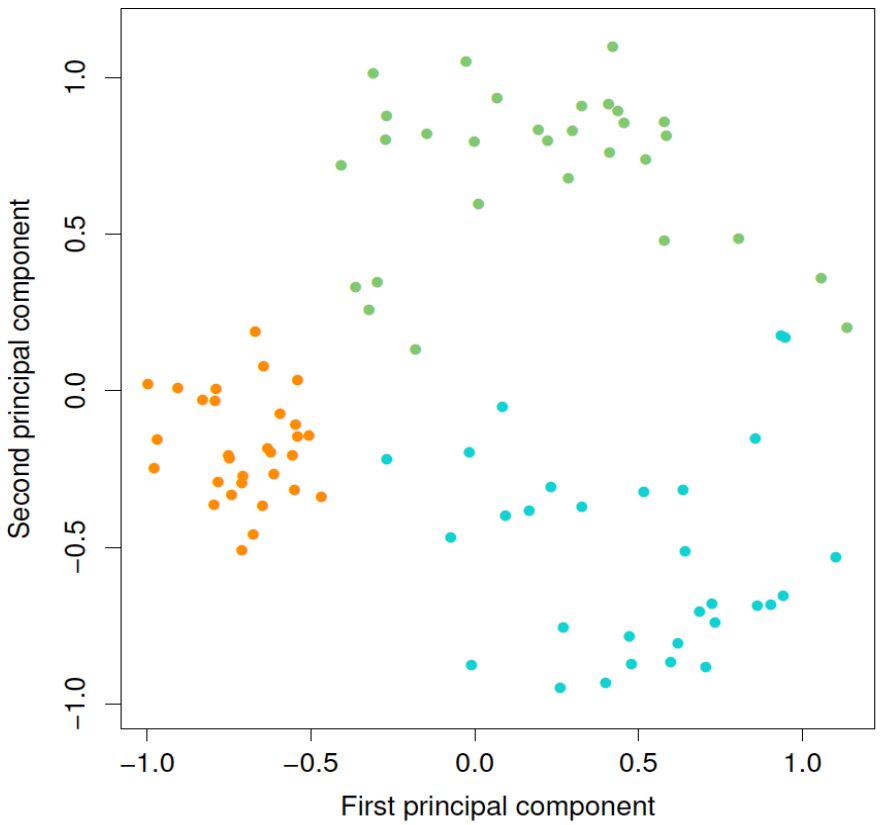
The first two principal component loadings for the `USArrests` data.

	PC1	PC2
Murder	0.5358995	-0.4181809
Assault	0.5831836	-0.1879856
UrbanPop	0.2781909	0.8728062
Rape	0.5434321	0.1673186

Another Interpretation

PCA find the hyperplane closest to the observations.





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Another Interpretation

- The first principal component loading vector has a very special property: it defines the line in p-dimensional space that is closest to the n observations (using average squared Euclidean distance as a measure of closeness).
- The notion of principal components as the dimensions that are closest to the n observations extends beyond just the first principal component.
- For instance, the first two principal components of a data set span the plane that is closest to the n observations, in terms of average squared Euclidean distance.

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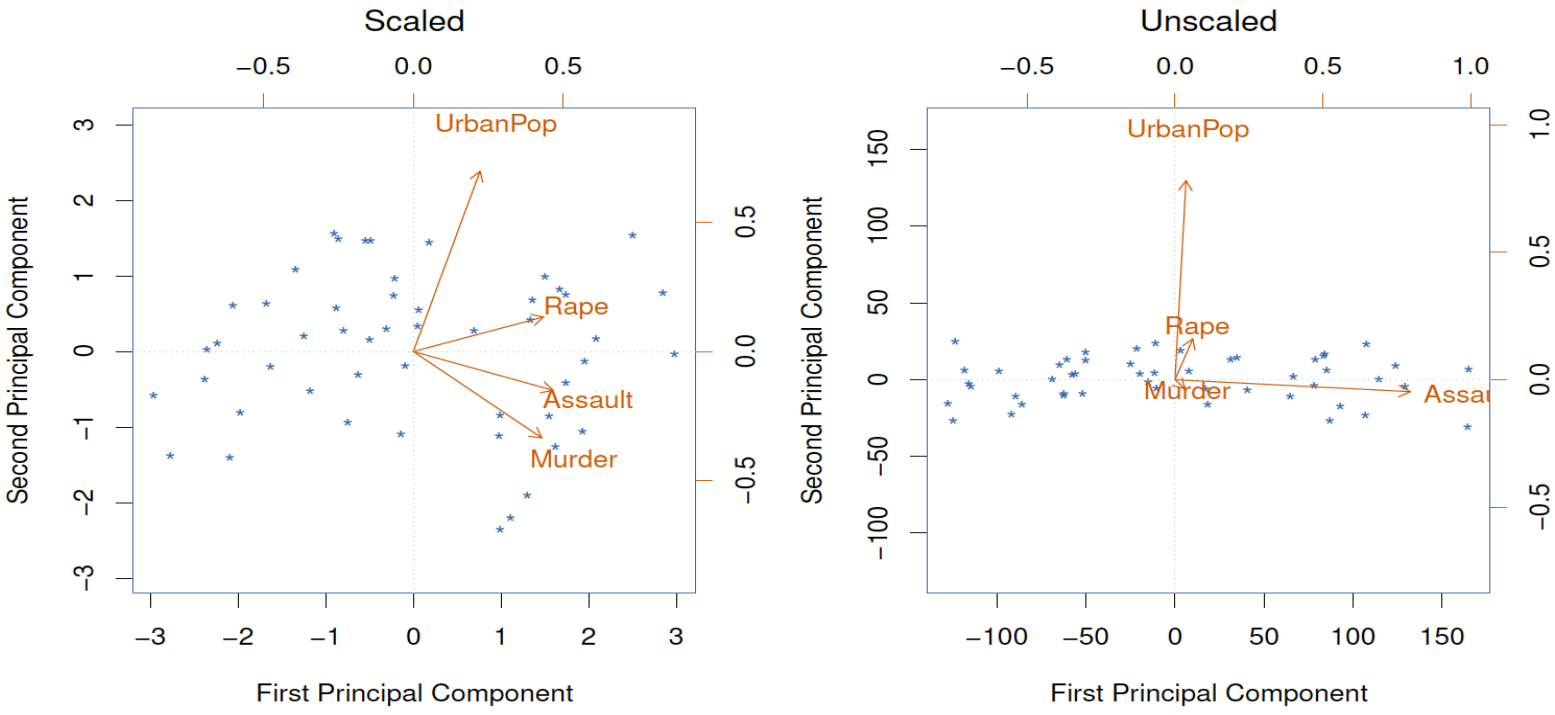
Scaling of the variables matters

- If the variables are in different units, scaling each to have standard deviation equal to one is recommended.
- If they are in the same units, you might or might not scale the variables.



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Proportion Variance Explained

- To understand the strength of each component, we are interested in knowing the proportion of variance explained (PVE) by each one.
- The total variance present in a data set (assuming that the variables have been centered to have mean zero) is defined as

$$\sum_{j=1}^{p} \mathrm{Var}(X_j) = \sum_{j=1}^{p} \frac{1}{n} \sum_{i=1}^{n} x_{ij}^2,$$

and the variance explained by the mth principal component is

$$\mathrm{Var}(Z_m) = \frac{1}{n} \sum_{i=1}^{n} z_{im}^2.$$

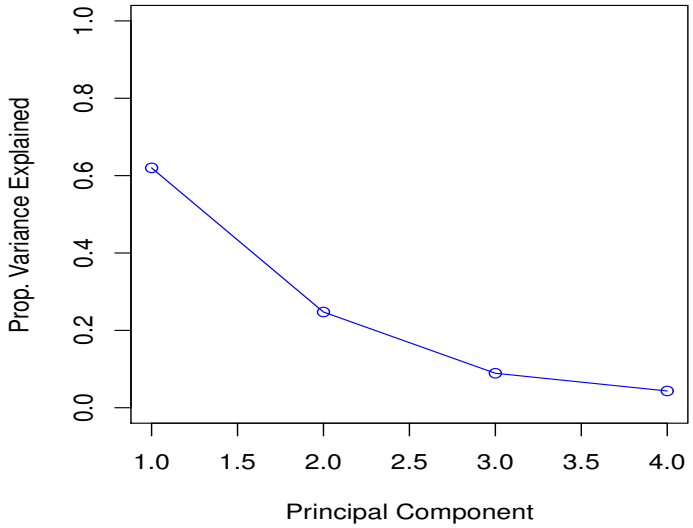
- It can be shown that $\sum_{j=1}^{p} \mathrm{Var}(X_j) = \sum_{m=1}^{M} \mathrm{Var}(Z_m)$ with $M = \min(n-1, p)$.

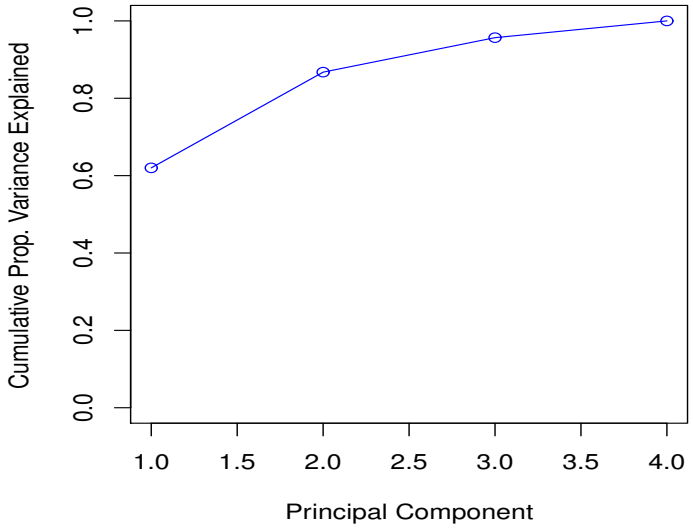
Proportion Variance Explained

- Therefore, the PVE of the mth principal component is given by the positive quantity between 0 and 1

$$\frac{\sum_{i=1}^{n} z_{im}^2}{\sum_{j=1}^{p}\sum_{i=1}^{n} x_{ij}^2} = \frac{\lambda_m^2}{\sum_{m=1}^{M} \lambda_m^2},$$

  where λ_m is the mth largest singular value of $\boldsymbol{X}$.
- The PVEs sum to one. We sometimes display the cumulative PVEs.





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Number of PCA

- If we use principal components as a summary of our data, how many components are sufficient?
- No simple answer to this question, as cross-validation is not available for this purpose. Why not?
- When could we use cross-validation to select the number of components?
- the scree plot on the previous slide can be used as a guide: we look for an elbow.

PCA vs Clustering

- PCA looks for a low-dimensional representation of the observations that explains a good fraction of the variance.
- Clustering looks for homogeneous subgroups among the observations.

Summary and Remark

- Example
- PVE
- Read textbook Chapter 10 and code